

Maya Winland-Gaetz

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Education	Bachelor of Engineering (BEng), In Progress, GPA 8.58/9 Expected Sept 2028 Electrical Engineering University of Victoria, Victoria, BC, Canada PhD Student, (Incomplete), GPA 11.0/12 September 2013 - May 2019 McMaster University, Hamilton, ON, Canada Bachelor of Science (BSc), GPA 3.57/4 September 2008 - April 2013 Specialist Environmental Toxicology, Maj. Ecology & Evolutionary Biology University of Toronto, Toronto, ON, Canada
Competencies	Technical • Drafted, simulated, and am currently in fabrication for several electronics projects using KiCAD, LTSpice, and Altium Designer. • Developed several applications in Python and C/C++ for personal use, and am currently learning Rust. • 90WPM typing speed with a keen eye for typographical and grammatical errors, using both Microsoft Office and LaTeX. • Prepared and analyzed sensitive geochemical samples in a laboratory environment, implementing state-of-the-art instruments while observing rigorous safety protocols. Core • Maintained efficiency while under pressure during the annual Starbucks Holiday Launch, providing a high standard of service for eager early-morning customers. • Communicated clearly and fluently in presentations and public speaking, earning an award at an academic conference for a 15 minute talk on geochemical sampler design.
Professional Experience	Electrical Engineering Co-op September 2025 - December 2025 Ocean Networks Canada, Sidney, BC, Canada • Developed an integrated data capture, storage, and graphing application using Python (with PyQt) and SQLite, which directly contributed to the detection and resolution of several mission-critical bugs prior to station deployment. • Compiled an exhaustive feasibility study for a 100VDC power distribution system redesign to be implemented in the Pacific Ocean Neutrino Experiment, including extensive component research, circuit design and multi-board modelling in Altium Designer, and transient analysis using LTSpice. • Utilized a number of industrial communication protocols in my software and firmware projects for interfacing with in-house and proprietary devices, including SPI, Telnet, RS-485, and MQTT. • Prepared comprehensive reports and documentation on all the above projects, to ensure future co-ops/employees can build upon my work. Teaching Assistant September 2013 - April 2018 McMaster University, Hamilton, ON, Canada

- Planned and conducted lab sessions and tutorials, using previous experience and WHMIS training in chemistry lab safety and analytical protocols, to minimize accidents, control spills and ensure a safe learning environment.
- Graded and annotated student papers and projects on 7-day deadlines, using rigorous organization and uniform standards, to ensure course leaders could maintain a strict schedule.

Projects

University of Victoria Satellite Design Club

January 2024 - Present

- Researched satellite radiocommunication design principles and protocols to guide the development of the telemetry systems for future club projects.
- Responsible for the design and assembly of the Telemetry, Tracking & Command (TT&C) boards and evaluation modules for the SLIP payload project, a partnership with UVic Rocketry.
- Hand-soldered 7mm QFN and 0603/0402-sized components for the board using a hot air rework station and manual soldering iron to assemble the first iteration of transceiver evaluation boards for testing.

First Year Engineering Design Project

January 2024 - April 2024

- Conceived and constructed a physical prototype and associated webapp for a smart shower heater as part of a class project. The product could sense the proximity of a user and automatically adjust the water temperature using a hypothetical inline water heater to a desired temperature, configurable through a WiFi-enabled app.
- Went beyond the expectations of the course by building transistor amplifiers to facilitate proximity detection with inexpensive IR sensors, and improved the TA-provided netcode to ensure a stable connection.
- Worked in a group of 3, wherein the circuit design and netcode improvements were my chief responsibilities.

Demonstration video available upon request.

Personal

April 2020-Present

- Currently building a trigger signal sequencer as part of a larger project to construct a music synthesizer, utilizing RP2040 hardware timers and digital logic ICs to output a 4-channel beat structure with a custom LED matrix.
- Programmed and assembled a climate and light level sensor for personal use, incorporating an AtMega328P microcontroller and a scrolling 16x8 LED display.
- Programmed an app in C++ to view a directory of life drawing references at regular intervals, to aid in life drawing study during the pandemic.

Awards and Certifications

- Amateur Radio Operator Certification December 2024
Advanced, Basic with Honours (VA7MWX)
- Best Student Oral Presentation, worth \$750 (*Sudbury Mining and Environment Int'l Conference*) June 2015
- WHMIS Yearly Recertification 2008-2022

Other Interests

- Bicycling, with the goal of getting from the University of Victoria to Leechtown and back over a two day trip.
- Classical piano, playing Beethoven's later Sonatas in particular.

References Available Upon Request